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Cerebral Palsy

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SUMMARY PEARLS

- Cerebral palsy (CP) is a spectrum disorder most commonly caused by prematurity related to ischemic and hemorrhagic injury along the periventricular tracts. It is defined by a nonprogressive disturbance of the brain that results in at least abnormalities of movement and posture.
- CP is commonly classified by body part(s) affected, type of movement or tone anomaly, and degree of function.
- The Prechtl General Movement Assessment and Hammersmith Infant Neurological Examination are tools used to predict and diagnose CP in neonates before 6 months of age; they are used best in combination with brain imaging.
- Positive prognostic indicators for function after CP are sitting independently by age 2 years, having less than three primitive reflexes by 24 months, having high visual and cognitive function, and lack of epilepsy.
- Tone management is a hallmark aspect of CP management and can include the use of therapy and bracing, enteral medications, injections, and surgical interventions.
- Hip subluxation can be a common progressive disorder in CP, occurring at higher rates in patients with higher Gross Motor Function Classification System scores.
- Neuromuscular scoliosis is a common progressive disorder in CP, which occurs at higher rates in individuals who are more severely affected.
- Physiatrists should work with individuals with CP and their families during adolescence as they transition to adulthood and be aware of their higher risk of stroke, myelopathy, loss of mobility, dementia, pain, fatigue, sleep disorders, and mood disorders.

Background

The definition of cerebral palsy (CP) has constantly evolved since the 1800s and has involved some of the most influential medical minds in history, including William Little, Sir William Osler, and Sigmund Freud. Advances in medicine and worldwide communication have led to the current definition: “Cerebral palsy describes a group of permanent disorders of the development of movement and posture, causing activity limitation, which are attributed to nonprogressive disturbances that occurred in the developing fetal or infant brain. The motor disorders of CP are often accompanied by disturbances of sensation, perception, cognition, communication, and behavior, by epilepsy, and by secondary musculoskeletal problems.”¹⁰³ This definition excludes a number of disease processes that may present similarly, but either progress (i.e., genetic or metabolic diseases), have a primarily peripheral etiology (i.e., neuromuscular or peripheral nerve injury), or occur after the central nervous system has fully developed (i.e., traumatic injuries or oncologic manifestations). It serves to remind us that the term CP reflects much more than a musculoskeletal process; it reaches to the function of a person. For example, each of the following cases is a possible presentation of a child with CP:

AB is a 10-year-old, left-handed female who was born at 36 weeks of gestation and whose brain imaging showed a prenatal stroke

on the left. She has difficulty using her right arm and leg because of right spastic hemiplegia for which she receives botulinum toxin injections three times a year. She uses an ankle-foot orthosis to run and play soccer with her friends. She is in regular classes at school and has no problem seeing, hearing, eating, talking, or socializing. She does note that she gets a “little upset” because she cannot keep up with her friends on the playground.

CD is a 10-year-old male who was born at 32 weeks of gestation and whose brain imaging showed significant periventricular leukomalacia. He has difficulty using both of his legs to walk because of spastic diplegia, for which he receives botulinum toxin injections three times a year and takes an oral medication three times a day. He requires bilateral ankle-foot orthoses and a reverse walker for ambulation in the community; at home, he crawls to get around. In addition, he has difficulty coordinating fine motor tasks with either hand but can put on his clothes with assistance for buttons and zippers. He requires accommodations at school for his mobility and dysarthria, needs glasses, and has difficulty being understood when talking. He has no problems with hearing or eating. He expresses that he wants other “little boys at school to play with him.”

EF is a 10-year-old male who was born at 24 weeks of gestation and whose brain imaging showed diffuse encephalomalacia

and hydrocephalus that required shunting. He has difficulty using all of his limbs because of spastic quadriplegia, for which he receives botulinum injections four times a year and takes two oral medications three times a day (one for spasticity and one for dystonia). He is mobile only in a customized wheelchair and requires bilateral ankle-foot orthoses and bilateral resting hand splints for positioning. He has numerous other impairments, including scoliosis, dysphagia, nutritional compromise requiring a gastrostomy tube for feeding and hydration, constipation, and significant intellectual disability.

Epidemiology

CP is the most common cause of disability in developed countries. The direct and indirect costs associated with CP are three to four times that of a neurotypical child.⁵¹ The global birth prevalence of CP ranges from 1.6 to 3.4 per 1000 live births. Low- and middle-income countries have reported birth prevalence in the higher range, whereas high-income countries have observed a declining trend in prevalence since the mid-2000s. This difference in international CP birth prevalence likely reflects the widely varied access to pre- and perinatal care.⁶³ When the overall prevalence is examined more closely, dramatic increases are seen in children with very low birth weight (59.6 per 1000 live births) and very early gestation (111.8 per 1000 live births).⁸⁶ Research has found the germinal matrix surrounding the ventricles is at increased risk of ischemic or hemorrhagic injury caused by the immature vasculature and fluctuations in cerebral blood flow.⁴

Etiology/Risk Factors

Many times, the etiology of the brain injury associated with the clinical findings of CP is clear, as in a history of placental

abruption with significant cardiorespiratory compromise causing global hypoxic ischemic injury resulting in spastic quadriplegia. However, it may not be as clear in a near-term infant with an uncomplicated hospital course who is developmentally delayed. Data from CP registries reveal that a larger portion of children with CP have a multifactorial etiology from a mixture of known risk factors.⁷⁵ There are five major categories of risk factors associated with the diagnosis of CP (Table 48.1).

There is increasing evidence for genomic variants contributing to CP etiology. Advances in sequencing technologies have allowed rapid and cost-effective genome or exome sequencing, revealing causative gene mutations in CP syndromes (GAD1, KANK1, or adaptor protein complex-4), novel gene mutation (ZC4H2), or pathologic single-nucleotide variants.^{43,72} Studies using chromosomal microarray analysis uncovered rare gene-level copy number variants associated with CP.^{87,114,134,137} A systematic review and meta-analysis by Gonzalez-Mantilla et al. demonstrated a 31.1% genetic diagnostic yield in CP in pediatric and adult patients, similar to rates of other neurodevelopmental disorders for which exome sequencing is standard of care. The findings provided evidence in support of the inclusion of CP in exome sequencing as part of the standard of care in diagnostic evaluation of patients with neurodevelopmental disorders.³⁵

Classification

One classification system for children with CP uses the topography or distribution of affected limbs. In this system, *hemiplegic CP* (Fig. 48.1) refers to a unilateral upper and lower limb distribution. It is common in term and near-term infants and associated with focal brain pathology (unilateral strokes, asymmetric periventricular leukomalacia [PVL], or congenital malformations). *Quadriplegic CP* or *tetraplegic CP* (Fig. 48.2) refers to a bilateral upper and lower limb distribution and is associated with diffuse brain pathology (cortical, subcortical, and intraventricular lesions) (Table 48.2).

TABLE 48.1 Risk Factors Associated With Cerebral Palsy

Preconception	Antenatal	Intrapartum	Neonatal	Postnatal
Maternal seizures	Birth defects	Birth hypoxia	Seizures	Stroke
Intellectual disability	Small for gestational age	Meconium staining	Respiratory distress	Abusive head trauma
Thyroid disease (hyper or hypo)	Low birth weight	Meconium aspiration	Hypoglycemia	Bacterial meningitis
History of stillbirth or neonatal death	Placental abnormalities	Abnormal duration of labor	Infections	Motor vehicle crashes ¹⁰
Maternal age >40 years	Maternal disease during pregnancy (respiratory, heart, seizures, incompetent cervix)	Fetal presentation ³⁴	Jaundice ³⁴	
Low socioeconomic status ³⁴	Abnormalities in fluid volume			
	Maternal bleeding in second and third trimesters			
	Hypertension			
	Preeclampsia			
	Chorioamnionitis ³⁴			



• Fig. 48.1 Child with spastic hemiplegia.



• Fig. 48.2 Child with spastic quadriplegia.

It is also common in term and near-term infants. *Diplegic CP* (Fig. 48.3) is a term that refers to a bilateral lower limb distribution and is most common in preterm infants. The associated brain pathology is PVL (Fig. 48.4).

The diagnosis can be further classified by the motor sign. Over the past 10 years, definitions for these motor signs have been detailed to provide reliable, valid terminology. Positive motor

TABLE 48.2 Grades of Intraventricular Hemorrhage in the Premature Brain

Grade	Hemorrhage
1	Isolated to the germinal matrix
2	With normal ventricular size
3	With ventricular dilation
4	With parenchymal hemorrhage

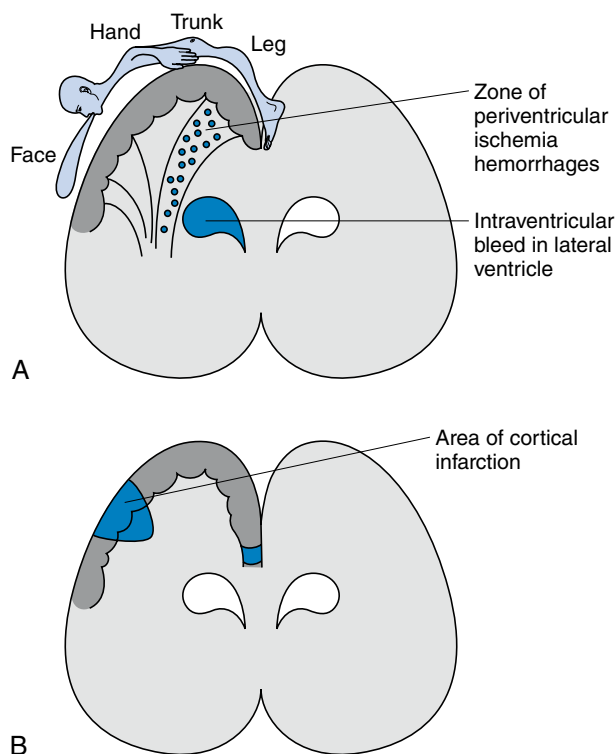


• Fig. 48.3 Child with spastic diplegia.

signs are defined as involuntary increases in frequency or magnitude of muscle activity. They include hypertonia (“examiner induced abnormally increased resistance to movement about a joint”)¹⁰⁹ and hyperkinesia (“unwanted, excess movement by the child”).¹⁰⁷ Hypertonia is further subdivided into spastic, dystonic, and rigid hypertonia. Hyperkinetic movements are divided into dystonia, chorea, athetosis, myoclonus, tremor, tics, and stereotypies. Negative motor signs are described as “lack of muscle activity or control” and encompass weakness, reduced selective motor control, ataxia, and apraxia (Box 48.1).¹⁰⁸

The push for standardized terminology has benefitted from the development of many CP registries across the globe. One important network, the Surveillance of Cerebral Palsy in Europe (SCPE), has standardized terminology and a reporting structure. Their terms include *unilateral spastic CP*, *bilateral spastic CP*, *dystonic CP*, *choreoathetoid CP*, *ataxic CP*, and *nonclassifiable*.¹⁴

It is important to note that, although one topographic distribution or motor sign may predominate, CP is commonly more complicated, and multiple motor signs and distributions may be seen in each patient. Another important aspect of describing the clinical findings associated with CP is functional. The Gross Motor Function Classification System (GMFCS) is a well-accepted and widely used classification system. It was revised in 2006 and has been shown to be valid and reliable. This system quantifies the typical



• **Fig. 48.4** Areas of brain damage causing diplegia vs. hemiplegia. (A) In spastic diplegia and periventricular leukomalacia, the leg is more affected than the hand and face. There is no cortical injury. (B) In spastic hemiplegia, the arm is often more affected than the leg. Because of cortical involvement, seizures and cognitive issues may occur.

• BOX 48.1 Classification of Motor Signs

Positive Motor Signs

Hypertonia

- Spasticity
- Dystonia
- Rigidity

Hyperkinesia

- Dystonia
- Chorea
- Athetosis
- Myoclonus
- Tremor, tics, and stereotypies

Negative Motor Signs

- Weakness
- Reduced selective motor control
- Ataxia
- Apraxia

performance in gross motor tasks (sitting, standing, walking, and using stairs) with mobility aids into five levels for five age groups. It has been used to develop motor development curves to prognosticate the age at which peak gross motor function is attained.^{92,104} Classification systems for upper extremity use (Manual Ability Classification System [MACS]),²⁷ communication (Communication Function Classification System [CFCSS]),⁴² eating/feeding (Eating and Drinking Ability Classification System),⁹⁴ and vision

(Visual Function Classification Scale)⁹⁸ have also been developed (Table 48.3).

In recognition that clinical findings in CP also include disturbances of sensation, perception, cognition, communication, and behavior, a multifaceted classification system was proposed in 2006. It recommended defining CP in terms of four components: motor abnormalities, accompanying impairments, anatomic and neuroimaging findings, and causation and timing.¹⁰³

In 2007, the World Health Organization (WHO), in collaboration with many other organizations, published the *International Classification of Functioning, Disability and Health for Children and Youth* (ICF-CY). This was an expansion of the original ICF for children from birth to 18 years of age to specifically address the developing child, the developmental delay, and the child in the context of the family. The ICF-CY provides internationally accepted terminology and classifications that focus on the relationship between structure and function and moves from a medical model of disability to a biopsychosocial model of care (Fig. 48.5).^{112,133} In this model, biopsychosocial function is broken down into five categories: body function, body structure, activity and participation, environmental factors, and personal factors.

Diagnosis

The term CP does not reflect a specific etiology; it is a description of clinical findings. There are many disease processes that can present as CP. The initial evaluation of a child with suspected CP should determine the diagnosis (if possible), decide which tests are needed to make the diagnosis, determine the etiology of the symptoms, determine comorbidities, and develop treatment options. Early diagnosis or identification of infants at high risk of CP allows early intervention that may potentially promote cognitive and motor development. Several assessment tools have been utilized to predict CP risk in infants.

The Prechtl General Movement Assessment (GMA) is based on observation of infants' spontaneous movements, known as general movements (GMs), that persist until 4 to 5 months of corrected age. The GMs of an infant comprise of a "writhing" phase from early fetal life up to 6 to 9 weeks of corrected age, then "fidgety" phase occurs 6 to 9 weeks of corrected age through 20 weeks of corrected age. Abnormal writhing movements are characterized as poor repertoire, cramped synchronized, or chaotic, whereas fidgety movements are classified as normal, abnormal, sporadic, or absent.²⁶ Although false positives are high with writhing GMs with a high sensitivity of 93% to 100% but low specificity of 40% to 59%, studies have demonstrated sensitivity of 95% to 100% and specificity of 89% to 98% for fidgety GMs for predicting CP at 2 years of age.^{56,117} The GMA is a more predictive tool during the fidgety period, with the most accurate time for assessment at 12 to 14 weeks of corrected age.⁸⁴

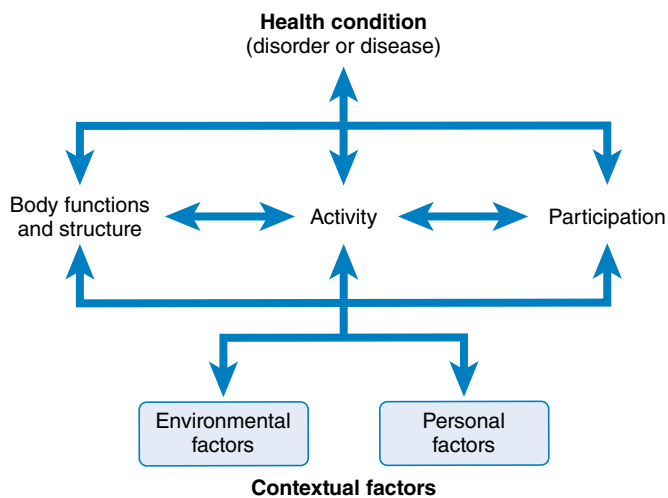
The Hammersmith Infant Neurological Examination (HINE) is a validated tool using scorable standardized neurologic examination to identify infants between 2 and 24 months of age who are at risk for CP. The examination includes 26 items, evaluating infants' cranial nerves, posture, movement, tone, and reflexes.¹⁰² A widely accepted cutoff HINE score of less than 57 of a maximum 78 at 3 months of corrected age is predictive of CP with a high sensitivity of 90% to 96% and specificity of 85% to 87%.^{40,82,102} The Hammersmith Neonatal Neurological Examination (HNNE) is another screening tool to detect neurologic differences in preterm and term-born infants up to 3 months

TABLE 48.3**Function Classification Scales: GMFCS, MACS, CFCS, EDACS, VFCS^{94,98}**

CLASSIFICATION SYSTEM					
Level	GMFCS	MACS	CFCS	EDACS	VFCS
I	Walks without limitations	Handles objects easily and successfully	Effective sender and receiver	Eats and drinks safely and efficiently	Uses visual function easily and successfully
II	Walks with limitations	Handles most objects with reduced quality and/or speed	Effective but slow-paced sender and receiver	Eats and drinks safely with limitations to efficiency	Uses visual function successfully but needs compensatory strategies
III	Walks using a handheld mobility device	Handles objects with difficulty; needs help to prepare and/or modify activities	Effective sender and receiver with familiar partners	Eats and drinks with some limitations to safety and efficiency	Uses visual function with some adaptations
IV	Self-mobility with limitations; may use powered mobility	Handles a limited selection of objects in adapted situations	Inconsistent sender and receiver with familiar partners	Eats and drinks with significant limitations to safety	Uses visual function in very adapted environments
V	Transported in a manual wheelchair	Does not handle objects	Seldom effective sender and receiver with familiar partners	Unable to eat or drink safely, feeding tube	Does not use visual function even in very adapted environments

CFCS, Communication Function Classification System; GMFCS, Gross Motor Function Classification System—Expanded and Revised; MACS, Manual Ability Classification System; EDACS, Eating and Drinking Ability Classification System; VFCS, Visual Function Classification System.

From Pålman M, Gillberg C, Himmelmann K. Autism and attention-deficit/hyperactivity disorder in children with cerebral palsy: high prevalence rates in a population-based study. *Dev Med Child Neurol.* 2021;63(3):320-327.



• **Fig. 48.5** Interactions between the components of International Classification of Functioning. (Redrawn from World Health Organization. *ICF-CY, International Classification of Functioning, Disability, and Health: Children & Youth Version*. WHO; 2007.)

postterm age. The scored neurologic examination consists of 34 items across six domains (tone, tone patterns, reflexes, spontaneous movements, abnormal signs, and behavior) with a global optimality score of 34.^{23,25} An optimality score of 30.5 or more for term-born infants and 26 or more for preterm infants when HNNE is administered at term-equivalent age is considered normal.⁶⁶ In infants born very preterm (<31 weeks of gestational age), the HNNE provides moderate predictive accuracy for motor

outcome at 12 months of corrected age, with stronger predictive ability when administered earlier in the neonatal period (32 weeks of postmenstrual age) than at term-equivalent age.⁴⁵

A combination of GMA, HINE, and brain magnetic resonance imaging (MRI) is currently the recommended toolset for predicting CP with the highest predictive validity before 5 months of corrected age.^{10,40,82} After 5 months of corrected age, MRI, HINE, and the Developmental Assessment of Young Children (a parent report measure of adaptive behavior, cognitive, social-emotional, physical, and communication domains) are the most predictive tools for CP detection.⁸² To assist with early detection of CP or identification of high risk, Novak et al. published an algorithm of best available evidence pathways, consisting of clinical history and presentation, standardized assessment tools, and neuroimaging.⁸² Before commencing any examination of an individual with suspected CP, it is important to gauge the emotional environment and the family's understanding of the child's current medical issues. Although many families bring their child to be evaluated because they believe something is different or wrong, they can be overwhelmed and afraid that their fears will be confirmed.

History

A detailed history should include all basic attributes of a medical history, the individual's birth history, family history of blood clots or stroke, and any possible risk factors for CP (see earlier). The developmental history should include when and how a milestone was achieved. For example, a child may have crawled at 9 months, but there are different patterns of crawling (e.g., "commando crawling," in which the child pulls the body across the floor with

the arms; “bunny hopping,” which refers to using flexion at the hip and knees but no reciprocal motions below the waist). Each of these crawling patterns can be associated with increased muscle tone in the legs. If there is regression of milestones after attainment, this suggests a neurodegenerative process and may indicate that a metabolic or genetic disease workup is also needed. A functional history will assist in determining treatment and life goals. This should include discussing the individual’s mobility in the home and community setting and participation in activities of daily living and community activities.

Clinical Findings and Patterns

A thorough medical examination should include detailed neurologic, musculoskeletal, and functional components. Areas of weakness as well as tonal and movement abnormalities should be identified. A child may have multiple tonal abnormalities such as spasticity, dystonia, and truncal hypotonia. In addition to positive and negative motor signs, the persistence of primitive reflexes is a common finding in individuals with CP (Table 48.4).¹³⁵ Having individuals with suspected CP demonstrate the extent of their mobility, object handling, transfers, and communication will provide valuable information.

Neonatal Pattern

After a severe brain injury, neonates tend to be hypotonic and develop hypertonia over time. It can be difficult to elicit muscle stretch reflexes, and there are minimal milestones to evaluate. Milestones may be delayed in premature children until the age of 2 years. Repeat examinations should be completed regularly.

Diplegic Pattern

Hypertonic deficits are noted mainly in bilateral lower limbs, though upper limbs may be involved. Scissoring of bilateral lower limbs and talipes equinovarus deformities predominate. If ambulatory, then toe walking, toe walking with knees flexed, or crouching tends to be present. Loss of range of motion (ROM) and hyperreflexia is present in the lower limbs. In a recent review, 72% of children with this pattern had neuroimaging that revealed changes on MRI.⁹⁹ Among children with CP, 38% will have this pattern.⁸¹

Hemiplegic Pattern

Hypertonic focal deficits, early hand lateralization, loss of ROM in affected limbs, and asymmetric limb use are associated with

TABLE 48.4 Eliciting Primitive Reflexes

Reflex	Position	Method	Response	Age at Disappearance
Palmar grasp	Supine	Placing the index finger in the palm of the infant	Flexion of fingers, fist-making	6 mo
Plantar grasp	Supine	Pressing a thumb against the sole just behind the toes in the foot	Flexion of toes	15 mo
Galant	Prone	Scratching the skin of the infant's back from the shoulder downward, 2–3 cm lateral to the spinous processes	Incurvation of the trunk, with the concavity on the stimulated side	4 mo
Asymmetric tonic neck reflex	Supine	Rotation of the infant's head to one side for 15 sec	Extension of the extremities on the chin side and flexion of those on the occipital side	3 mo
Suprapubic extensor	Supine	Pressing the skin over the pubic bone with the fingers	Reflex extension of both lower extremities, with adduction and internal rotation into talipes equinus	4 wk
Crossed extensor	Supine	Passive total flexion of one lower extremity	Extension of the other lower limb with adduction and internal rotation into talipes equinus	6 wk
Rossolimo	Supine	Light tapping of the second to fourth toes at their plantar surface	Tonic flexion of the toes at the first metacarpophalangeal joint	4 wk
Heel	Supine	Tapping on the heel with a hammer, with the hip and knee joint flexed, and the ankle joint in neutral position	Rapid reflex extension of the lower extremity in question	3 wk
Moro	Supine	Sudden head extension produced by a light drop of the head	Abduction followed by adduction and flexion of upper extremities	6 mo
Babinski	Supine	Striking along the lateral aspect of the sole extending from the heel to the head of the fifth metatarsal	Combined extensor response: simultaneous dorsiflexion of the great toe and fanning of the remaining toes	Presence always abnormal

From Zafeiriou DI. Primitive reflexes and postural reactions in the neurodevelopmental examination. *Pediatr Neurol.* 2004;31:1-8.

hemiplegic CP. Persistent primitive reflexes may be present. In a recent review, 84% of children with this pattern had neuroimaging that revealed MRI changes.⁹⁹ Among children with CP, 39% will have this pattern.⁸¹

Quadriplegic Pattern

There is diffuse spastic hypertonicity with truncal dystonia, persistent primitive reflexes, intellectual disability, hyperreflexia, and loss of ROM in all affected limbs. In a recent review, 92% of children with this pattern had neuroimaging that revealed MRI changes. Among children with CP, 23% will have this pattern.⁸¹ The clinical presentation will guide the needed workup and development of a comprehensive treatment plan. For example, a prenatal stroke could cause a hemiplegic pattern indicating the need for a coagulation workup and neuroimaging to evaluate for a brain malformation, as is found in 10% of children with this pattern.⁹⁹ Although diffuse hypotonia may be a precursor to hypertonia, persistent hypotonia is often associated with genetic or metabolic syndromes.

Functional Prognosis

In the 1970s, Molnar et al. laid the foundation for the future development of motor curves based on developmental milestone achievement. Their work showed that sitting independently by the age of 2 years or having less than three primitive reflexes present by 18 to 24 months was a positive predictor of ambulation, although not sitting independently by 4 years was a negative predictor.⁶⁹ A systematic review and meta-analysis of observational studies demonstrated that other positive predictors of ambulation include absence of visual impairment, absence of intellectual disability, and absence of epilepsy or seizures, in addition to sitting independently by age of 2 years.⁵² Later work by Molnar et al. demonstrated that over 50% of all children with CP will ambulate, and they further divided this by topographic classification (80%–90% diplegia, 50% quadriplegia, 75% dyskinesia).⁷⁰ In 2008, these findings were replicated in a large European population of children with CP.⁶

Moving from the prediction of ambulation to the prediction of longitudinal gross motor function in children with CP, Rosenbaum et al. published motor development curves based on an individual's GMFCS level in 2002.¹⁰⁴ These curves helped transition the diagnosis of CP from a disease of childhood to a chronic disease. They have been validated, and their stability has been shown since their initial publication. Interestingly, children with GMFCS levels I and V often remained at these levels, while children with GMFCS levels II, III, and IV needed reclassification over time.¹⁰⁴ These findings were replicated in a large cohort in Europe.⁸³ Reclassification of MACS and CFCS is also more likely to occur in younger children (age ≤ 4 years), particularly for those whose initial GMFCS level was II through IV, initial MACS level was III, or initial CFCS was II through V.⁹¹ As the ICF-CY model continues to modify our paradigm of health, efforts to assess children with CP in the domain of activity and participation will likely increase. Most recently, the SCPE has been working to develop development curves for self-care capabilities,⁵⁴ social participation,¹²³ and health-related quality of life.¹²³

Medical Management

The spectrum of associated medical conditions affecting individuals with CP is broad. In the last consensus definition of CP, there

is a greater emphasis on recognizing associated impairments and functional limitations as well as coexisting medical conditions, aside from the primary abnormalities in movement and posture.¹⁰³ Function, participation, and quality of life are often influenced by these comorbidities. It is intuitive that the more severe the motor disorder or higher GMFCS level, the greater the likelihood that the child will have multiple or more severe medical comorbidities. Two exceptions to this correlate are pain occurring in all levels of disability and across the lifespan and behavior disorders seen more commonly in cases of milder motor disability.⁷⁹ Etiologies of CP and their anatomic correlates may also predict these associated conditions. Some associated conditions, such as seizures, are clearly attributed to a disturbance at the brain level, but others may be caused by a combination of abnormalities in neural control and sensorimotor deficits. Though described as a nonprogressive condition, aging with CP is associated with declining health and function status across neurologic and nonneurologic systems.¹³⁰ Whitney et al. described that among 16,728 adults with CP, the most common comorbidity combinations were cardiorespiratory diseases, dysphagia, and fluid/electrolyte disorders; metabolic disorders (e.g., diabetes, renal disease, hypertension); neurologic-related disorders (e.g., dementia, cerebrovascular disease); gastrointestinal issues; and orthopedic-related disorders.¹³⁰

Feeding, Growth, Nutrition, and Gastrointestinal

Dysphagia is common in patients with CP. Impaired oropharyngeal strength and coordination place a child at risk for not meeting caloric and fluid requirements because of feeding inefficiency. Serious medical complications, such as malnutrition and an increased risk of aspiration events, may occur. Caregivers may share that this aspect of the child's care is burdensome, with stress increasing when the child needs considerable time to eat, and activities revolve around feeding schedules. Frequently, children will struggle to gain weight only to lose it quickly during an illness. It is important to remember that some individuals have silent aspiration and may not generate a protective cough; therefore workup for aspiration events should include a formal swallow study. Early on, children should receive nutritional counseling and supplemental caloric and fluid intake if necessary. Parents may resist placement of gastrostomy tubes, especially if the child swallows safely, even though feedings may require increased time and aggressive positioning strategies. Oral hygiene may be compromised, and children may have excessive sialorrhea, which can contribute to aspiration events and cause secondary skin irritation. Growth and nutrition may be compromised because of limited intake of key nutrients, impaired absorption, and endocrine abnormalities. When malnourished, children are at risk for growth disturbances, increased infection rates, skin breakdown, osteopenia, and ultimately decreased life expectancy.⁵⁹

Usual tools for weight and nutrition management are often not reliable in children with CP. Published guidelines for nutritional surveillance in children with neurodisability recommend use of triceps skinfold (TSF), mid-upper arm circumference (MUAC), and the derived measure of arm muscle area (AMA), all in addition to weight for age.¹¹⁹ Red flags for undernutrition in children with neurologic impairment include physical signs of undernutrition; weight for age Z-score below 2; TSF, MUAC, or AMA below the 10th percentile; and faltering weight over time.¹¹⁹ The North American Growth in Cerebral Palsy Project developed CP-specific growth curves.^{48,101} These growth charts highlight increased mortality rates for children GMFCS levels III

through V, with weight below the 20th percentile. Gastroesophageal reflux disease (GERD) and constipation are common in CP, may produce discomfort, and exacerbate hypertonicity. In more severe cases, children may have dysmotility with delayed gastric emptying and esophagitis from medically refractory GERD.

Pulmonary

Lung disease is an important cause of morbidity and mortality in CP. The leading cause of death in children with CP is respiratory illness. For adults with CP, death from respiratory illness is 14 times higher than for their peers with no disability.¹¹ Cumulative injury to lung architecture may occur because of repeated aspiration events, infections, decreased mucociliary clearance, kyphoscoliosis, and airway obstruction. Nine risk factors that place a child with CP at high risk for having at least one hospital admission for chest infections in the next 5 years include GMFCS level V (difficulty controlling head and body posture in most positions), at least one hospital admission for respiratory illness in the past year, oropharyngeal dysphagia, seizures, frequent respiratory symptoms, GERD, at least two courses of antibiotics for chest infections in the past year, mealtime respiratory symptoms, and snoring.³⁴ Evaluation recommendations outlined in a consensus statement focusing on prevention and management of respiratory illnesses in individuals with CP underscored need for detailed evaluation, including medical and mealtime history, physical exam/upper airway evaluation, clinical and video fluoroscopy swallow assessments, and (when indicated) computed tomography of the chest. It stresses the need to manage risk factors of aspiration, including GERD and drooling. Some of the strategies discussed include modifying feeding consistencies, positioning, equipment to optimize feeding, treatment of GERD, feeding tubes, and control of sialorrhea with anticholinergic medications, botulinum injections, and sometimes surgery. The need for optimizing airway clearance and chest wall mobility was underscored.³⁴ This can be achieved with suction or chest physiotherapy techniques. Devices such as cough-assist machine and vibration percussion vests can be used. Children with infections should be treated with tailored antibiotics based on sputum cultures. Some will require prophylactic antibiotic coverage. Preventative measures include immunizations because this population is at higher risk for infections and complications. Children with CP frequently have sleep-disordered breathing. A history of snoring or observed irregular breathing patterns with or without excessive daytime somnolence should prompt the practitioner to investigate sleep apnea, which if untreated can contribute to further morbidity. Apnea may be central, peripheral, or a combination of both. Many children have impaired sleep regulation and may not achieve restful sleep because of microaspiration and cough, altered light perception, medication side effects impacting natural sleep-wake cycles, and challenges in achieving comfortable positioning.³¹ Upper airway obstruction while awake and during sleep, with resulting hypoventilation, is an important contributor to respiratory morbidity and decreased quality of life in individuals with CP.⁹ Management can include surgery, noninvasive ventilation, or tracheostomy in very severe cases.⁹ Impaired lung function and restrictive lung disease are common in children with neuromuscular weakness and kyphoscoliosis. Early-onset kyphoscoliosis interferes with lung development by decreasing the number of alveoli. Scoliosis surgery may improve restrictive lung disease but must be considered carefully given risks associated with it and whether it will improve functioning and quality of life.

Neurologic Issues

The CP population has an increased risk for seizures. The prevalence of epilepsy in CP is estimated to be around 33% to 40%.^{16,121} Etiologies for seizures in CP may be related to vascular injury, brain malformations, or genetic and epileptic syndromes, among others. Seizures occur most frequently in children with a quadriparetic clinical presentation, GMFCS levels IV and V, term pregnancy, history of hypoxic-ischemic encephalopathy, and cognitive impairment.¹²¹ Children with CP and epilepsy have an increased number of hospitalizations and emergency department presentations compared to children with CP or epilepsy alone.¹²¹ Medication use for epilepsy management may be associated with cognitive dulling or other side effects. Some children have hyperkinetic movement disorders that mimic seizure activity and may require video electroencephalography monitoring to delineate between the two conditions. Weaning of seizure medications depends on the severity of the seizures and a period, usually 2 years, during which the child has been seizure free. Children with CP also have a higher incidence of intellectual disability; cognitive impairment is estimated to affect 50%.²² Nonverbal status and severe intellectual disability are seen more commonly in children with a quadriparetic clinical presentation. However, it is important to note that some children, especially with basal ganglia injury and dyskinesias, may have preservation of intellectual abilities despite severe dysarthria. It is also important to differentiate the cause of communication difficulties (cognitive dysfunction vs. hearing or vision limitations vs. dysarthria).¹ Children who are higher functioning and mobile may still have cognitive dysfunction and learning disabilities despite relative preservation of verbal language. Attention-deficit disorder and attention-deficit/hyperactivity disorder have a prevalence of 30% in CP.⁹⁰ Screening for impaired attention and executive functioning deficits is recommended. Visual perceptual deficits may also affect learning, despite a normal IQ.

Mood disorders, such as anxiety and depression, have a high prevalence in children with CP and can impact functioning and participation. Mood disorders are amplified in adulthood with an incidence of 28% vs. 12% compared to the general population. There is a considerable comorbidity with pain and fatigue in the CP adult population.¹³¹ Up to 30% of children and adolescents with CP have an autism diagnosis and other genetic or neurodevelopmental disorders.^{62,127} Individuals with both diagnoses often present with other comorbidities, such as intellectual disability, behavior, and sleep disorders. Depending on the location in the brain, some children will be at higher risk for vision and hearing deficits. Prematurity is associated with retinal damage and myopia. Overlap between CP, PVL, prematurity, and strabismus is known.⁴¹ Occipital lobe pathology may result in cortical visual impairment. Strabismus is important to recognize early to avoid amblyopia. Hearing deficits may be conductive and/or sensorineural in etiology. Children with CP caused by kernicterus or TORCH infections should be screened for sensorineural hearing loss.

Elimination

In CP, there is a high incidence of bladder and bowel dysfunction (BBD), with continence achieved later than peers. BBD includes daytime and nighttime enuresis, urinary urgency or frequency, encopresis (fecal soiling), chronic and functional constipation, and pain in the lower abdominal area and pelvis.⁵ BBD may result from impaired processing of sensory feedback from bladder and rectum, incoordination of pelvic floor muscles during elimination,

detrusor sphincter dyssynergia, and neurogenic detrusor overactivity. In CP, other factors affecting BBD and age at toilet training are cognitive ability, motor function level, and impairment of the autonomic system.⁵ Urinary retention can increase chances of urinary tract infections, and if a high intravesical pressure is generated, vesicoureteral reflux with resultant hydronephrosis and pelviectasis may occur. Clean intermittent catheterization is required when children are experiencing chronic urinary retention with urinary stasis. Anticholinergic treatments for dystonia or sialorrhea may trigger urinary retention. Urinary retention and urolithiasis can also produce discomfort and exacerbate hypertonicity.

Musculoskeletal Pain and Osteopenia

Pain in CP has been classified as nociceptive, nociplastic, neuropathic, or mixed neuropathic/nociplastic.¹¹¹ There is growing evidence for both inflammatory and centralized pain mechanisms.³² Recurrent musculoskeletal pain in children with CP is common, with a linear correlation with higher GMFCS level and age. Children who are less mobile are at increased risk for contracture and bony deformity with associated musculoskeletal pain. Common sites of musculoskeletal pain include the hips, spine, knee, and foot-ankle complex. A recent systematic review by Ostojic et al. on pharmacologic and surgical pain management in children and adolescents with CP found no scientific evidence to support the efficacy of interventions, such as different surgical procedures, botulinum neurotoxin type A (BoNT-A) injections, intrathecal baclofen (ITB) therapy, oral and inhaled drug therapies, and analgesic injections. Further research is recommended. The same applies to nonpharmacologic methods, such as physiotherapy, massage, environmental adjustments, and occupational therapy.⁸⁸

As people with CP age, they may develop secondary conditions, such as joint dislocations, contractures, osteoarthritis, stenosis, spondylolysis/lithesis, and neuropathy. These secondary conditions may result in chronic pain syndromes, which may be perpetuated by depression and other psychosocial factors.¹²⁶ A recent systematic review and meta-analysis looking at pain in adults with CP estimated the incidence of chronic pain in adults with CP to be 70% compared to 20% in general population. Chronic pain is experienced across all GMFCS levels and is associated with several adverse quality of life and functional outcomes. Lower back, hips, legs, feet, and neck were identified as the *most* painful locations in adults.¹¹¹ Chronic pain in adults with CP often goes underrecognized and undertreated. Patients with pain and CP are more likely to be exposed to opioids for longer periods of time than the non-CP population raising significant concerns about opioid use in CP.¹¹¹

Children with CP are at risk for osteopenia and related fragility fractures. Risk factors for osteoporosis include decreased weight bearing, use of anticonvulsants, malnutrition, and decreased sunlight exposure. Supplementation with calcium and vitamin D is routinely used to improve bone density, but the effects on prevention of fragility fractures are unclear. In those who are nonambulatory, it is recommended that their serum 25-hydroxyvitamin D level be 30 to 40 ng/mL.¹¹⁹ Bisphosphonates are effective in improving bone mineral density but are not without risk, and this intervention is usually reserved for children who have experienced a fragility fracture. Because the long-term side effects with use of bisphosphonates are unknown, additional research is required before they become routinely used.²⁹ Standing frames are used to facilitate weight bearing in children with CP who are nonambulators. A few studies

report a positive effect on bone density of standing for an average of 30 to 60 minutes a day without it being statistically significant.¹¹⁹

Therapeutic Management

The treatment of a person with CP requires a multidisciplinary team that understands the different presentations of this condition, the motor control abnormalities, and the primary and secondary conditions associated with its multisystem involvement. The team composition may vary according to the child's needs. It may include physiatrists, orthopedic surgeons, neurologists, neurosurgeons, developmental medicine specialists, physical therapists, occupational therapists, speech-language pathologists, orthotists, nutritionists, and social workers, among others. Additionally, it is important the family and child be active team members when setting priorities and goals within the context of the level of impairments and the child's age.

Because of the WHO ICF-CY recommendation that disease treatments be focused mainly on the individual's level of activity and participation, there has been a rising paradigm shift in the treatment of CP. The previous focus on alleviating the associated motor impairments has transitioned into discussions of how different treatment options will affect function.¹⁷ The goal of any therapeutic intervention is to encourage the child's maximal potential in motor, cognitive, and social realms.⁸⁵

Childhood Disabilities and Education

There are numerous special education laws ensuring that individuals of all ages receive educational services in the United States. [Table 48.5](#) summarizes pertinent legislation.⁴⁷

Overall, physiatrists should be familiar with three terms that have evolved. *Early intervention programs* are government-sponsored programs that are tasked to evaluate children at risk or with delays in one or more areas of function between the ages of 0 and 36 months.⁶⁷ Services include therapies, nutrition monitoring, care coordination, audiovisual, and social work, among others. After 36 months of age, children with disabilities may qualify to receive ongoing services through their local public school system. A *504 plan* ensures that appropriate classroom accommodations will be made for children who do not qualify for special education classes. An *individualized education program* ensures that classroom accommodations are provided, along with a document provided to the family with present level of functioning, short- and long-term goals, specific educational services needed, participation in regular educational programs, and assurance that the necessary services are available.⁴⁷

Therapy Interventions

There are many forms of therapy interventions. Therapists use a combination of approaches while tailoring their therapy program to each child's needs. Novak et al., in a 2019 systematic review, highlight what works and what does not to inform decision making and highlight areas for more research ([Fig. 48.6](#)).⁸⁰ The main goal of all therapies is to promote and facilitate development in all domains of function.³ The setting and frequency of formal therapy sessions vary. The most common therapy settings are school-based, outpatient models with a frequency of one to three times a week per discipline. Other models include an intensive "burst" of therapy through camps or day rehabilitation programs. Regardless of the setting, it is imperative that the family and child are

TABLE 48.5 Important Legislation and Laws for Children With Disabilities⁴⁷

Law/ Legislation	Summary
The Handicapped Children's Early Education Assistance Act of 1968 or Public Law (PL) 95-538	Provides for educational programs for young children with disabilities. Funds research institutes to study the behavioral, cognitive, and emotional functioning of children.
The Vocational Rehabilitation Act of 1973 (PL 93-112)	Provides for services for individuals with physical or mental handicaps that promoted independence and employability. Section 504 provides protection against discrimination of children who might not meet "special education" definitions but require classroom accommodations, such as extra time for test taking or frequent breaks.
The Education of All Handicapped Children Act of 1975 (PL 94-142)	Requires that no child, regardless of disability type or severity, can be excluded from a school education. Mandates that every child with a disability have an individualized educational plan that ensures appropriate accommodations, but also includes detailed plans and goals of education.
The Individuals with Disabilities Education Act (IDEA) of 1990 (PL 101-476)	Provides funding for infant and toddler early intervention programs. Replaces the word "handicapped" with "disabled." Ensures infants and toddlers with disabilities from birth to 2 years of age with developmental delay receive early intervention services. Requires that children ages 3–21 years receive special education and services.
The No Child Left Behind Act of 2002	Changes the distribution of federal funds from volume-based metrics to student performance-based metrics.
The Individuals with Disabilities Education Act (IDEA) revisions of 2004	Updates PL 101-476 to include a requirement that children with disabilities be prepared for further education, employment, and independent living.

From Hurwitz KA: A review of special education law, *Pediatr Neurol*. 2008;39:147–154, 2008.

included in the therapy plan to promote carryover and reinforcement in the home setting. Recently published evidence-based motor rehabilitation guidelines recommend first-line motor rehabilitation interventions for children and adolescents with CP should include aerobic and muscle strengthening exercises, gait training, an intensive exercise program such as Hand and Arm Bimanual Intensive Therapy (HABIT) or HABIT-Including Lower Extremities (HABIT-ILE), use of orthoses, physical activity and/or sport activities, and therapeutic education for the individual and family. For adults with CP, aerobic exercise, gait training, physical activity and/or sport activities, and therapeutic education for the individual and the family were proposed as first-line motor rehabilitation interventions.¹⁹

Stretching

Stretching is a component of therapy programs; its goal is to reduce the risk of contracture development as a result of muscle imbalances and hypertonicity. Sustained stretching can be achieved through the use of positioning devices, orthoses, and serial casting. A systematic review examining the effects of casting, either alone or in combination with BoNT-A, on equinus in children with CP revealed no strong and consistent evidence that combining casting and BoNT-A is superior to using either intervention alone.⁸ Data indicate that children tolerate casting better when it is applied 4 weeks after toxin injection.⁸⁰

Strengthening

Historically, exercise programs that included strengthening were contraindicated in CP. A systematic review of the effectiveness of strength-training programs in CP reported increased strength without any negative effects of increased spasticity or reduced

ROM. Indirectly, some studies have reported increased participation and improved self-esteem.²¹ Moreau et al., in a meta-analysis looking at rehabilitation interventions to improve gait speed, found strength training in CP, regardless of the type and devices used, does not result in clinically meaningful changes in gait speed. Preliminary work on resistance training for muscle power, such as velocity training, was shown to be an effective means of increasing gait speed compared with traditional strength training.⁷¹ High-intensity interval training in some studies has shown improved physical fitness, cardiometabolic risk biomarkers, mental health, and cognitive performance in children with a spectrum of developmental disabilities, including CP.⁹⁵

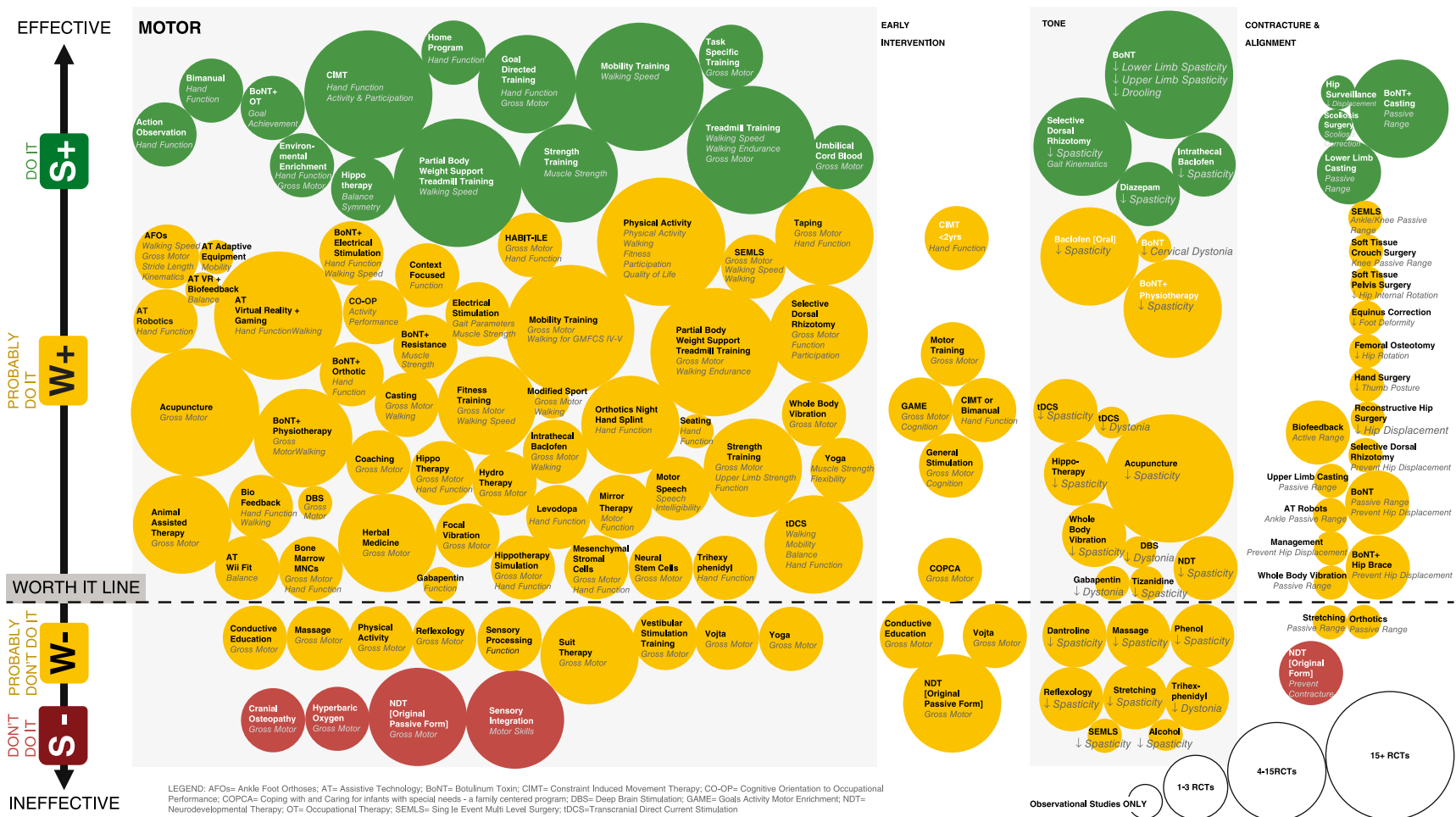
Aerobic Exercise

Historically, exercise programs that included aerobic exercise were contraindicated in CP. A Cochrane review looking at exercise intervention in CP concluded that aerobic exercise may result in a small improvement in gross motor function in the short and intermediate term and may improve physical activity without adverse events. Improvements in aerobic fitness have been described with 45 minutes four times a week of "high intensity" activities such as wheelchair sports, swimming, matt exercises, or cycling. None of the studies have reported outcomes representing the ICF model of increased activity and participation.^{100,105}

Constraint-Induced Movement Therapy

Constraint-induced movement therapy (CIMT) is a treatment for hemiparesis to improve motor function in the affected upper limb. In infants and children with hemiplegic or unilateral CP, the unaffected limb is restrained with a removable cast, typically for 3 weeks, and the child undergoes intensive, structured therapy in

• **Fig. 48.6** Evidence for motor, early intervention, tone and contracture management interventions. (From Novak I, Morgan C, Farley M, et al. State of the evidence traffic lights 2019: systematic review of interventions for preventing and treating children with cerebral palsy. *Curr Neurol Neurosci Rep.* 2020;20(2):3.)



addition to daily activities and play. A recent Cochrane review about CIMT concluded that the quality of evidence was low. For children with hemiplegic CP, there was some evidence that CIMT resulted in improved bimanual performance and unimanual capacity.⁴⁴

Hand Arm Bimanual Intensive Therapy and Hand Arm Bimanual Intensive Therapy—including Lower Extremity

HABIT and HABIT-ILE encompass bimanual upper extremity training with activities involving the trunk and lower extremities. It was initially designed for unilateral CP. However, recent studies have shown improvement in activities of daily living and dexterity in children with bilateral CP. These improvements were retained 6 months after initial treatment.³⁰

Functional Electrical Stimulation

Neuromuscular electrical stimulation (NMS) is the application of an electrical current of sufficient intensity to elicit muscle contraction. When applied during a functional activity, it is referred to as *functional electrical stimulation (FES)*. An example is FES-assisted cycling. There is evidence to support NMS and FES.^{53,80}

Robotic and Partial Body Weight Support Treadmill Training

The robotic and partial body weight support treadmill training (PBWSTT) forms of therapy involve a repetitive and task-specific approach to facilitate attainment of stepping and locomotion to achieve a more normalized gait pattern based on current theories of motor learning. The setting includes an overhead harness system used to support the child's body weight on a treadmill, while the therapist or a robot facilitates the kinematic, kinetic, and temporal features of walking. Although evidence of benefits from PBWSTT and robotic therapy has emerged in adults with stroke and spinal cord injury, several recent systematic reviews showed limited evidence to support the use of PBWSTT in children with CP. Despite reported improvements in gross motor function, functional status, walking performance, and gait parameters, statistical significance was not reached in most of the studies.^{74,96}

Neuromodulation

Transcutaneous spinal stimulation and transcranial motor stimulation are neuromodulation therapies proven to be safe, noninvasive, and well tolerated. They work by triggering neural network reorganization. There is emerging evidence to support their use in children and adolescents with CP.

Durable Medical Equipment

The goal in prescribing durable medical equipment (DME) for individuals with CP should focus on maximizing function, improving safety, and enabling independence using the ICF-CY model of health. Motor impairment is one of the top reasons for a mobility device in CP because exploration of the environment is imperative for cognitive development that leads to maximizing functional independence. Supportive or adaptive seating systems and standing frames can facilitate a developmentally appropriate upright posture, strengthening, flexibility across the lower extremities, weight bearing/bone density, upper limb function, communication, and feeding by freeing the child's hands to perform bimanual tasks, improving breath support, and optimizing the head and trunk position to facilitate a safe swallow.



• **Fig. 48.7** Child with spastic quadriplegia seated in a custom manual wheelchair.

Specialized seating devices are available for sitting on the floor or toilet, feeding, and bathing, as well as for incorporation in a mobility device.⁸⁵

Use of a wheelchair becomes applicable when a child either outgrows commercially available strollers or additional support is necessary. The goal of supportive seating is to provide an upright seated posture to facilitate interaction with the environment and minimize deforming forces secondary to postural abnormalities (Fig. 48.7). The clinician must consider both the seating goals and the family's needs. One decision unique to pediatrics is whether to use either a conventional wheelchair base or a stroller base for the mobility device. Power mobility can be pursued in children as young as 18 months old if they have fair motor control, no visual deficits, and good cognition.⁸⁵

For children with adequate head and trunk support, a gait trainer, walker, or crutches might help facilitate gait training and provide external support and improve upright posture. The gait trainer is a wheeled walker with a sling seat and various support options, which allows the child to propel the device without necessarily having a coordinated, reciprocal gait pattern. For the dependent child and adult, families might benefit from equipment to facilitate transfers and floor recovery, such as a mobile mechanical lift or overhead lift device.⁸⁵

The production of speech, language, and gesture for communication is also commonly affected in CP because of motor, sensory, and intellectual impairments. Various applications of technology have been used to maximize function. In augmentative communication, meaningful communication and expression of needs are facilitated with the use of computers, tablets, switch devices, signboards, and similar adaptive equipment.⁸⁵

Splinting and orthoses are commonly used in CP to manage spastic but flexible dynamic deformities of the extremities.

There are a variety of passive and dynamic splints. Orthoses must be tailored to the child's age, deformity, motor control, and tolerance. With lower limb orthoses, the clinician should clearly identify the gait deviation and goals to be addressed, with special consideration of ankle-foot alignment, ROM, and tone. Ankle-foot orthoses have been demonstrated to decrease the energy cost of walking in children with CP, compared with barefoot walking, and to improve gait parameters of stride length and velocity.⁸⁵

Management of Hypertonia

Hypertonia is one of the most prevalent comorbidities of CP that affect function. The most common forms observed in CP include spasticity, dystonia, and a combination of both.^{50,103} Spasticity is defined as velocity-dependent resistance, whereas dystonia is a more dynamic movement disorder, which involves an involuntary alteration in pattern of muscle activation during voluntary movement. It may fluctuate in presence and severity over time.¹⁰⁹ Learning to differentiate between them can guide goal setting and intervention. Hypertonia can be focal or generalized and can affect the limbs and trunk. If left untreated, it can interfere with function and development. Hypertonia can lead to secondary musculoskeletal complications such as joint contractures and dislocations. It can also be a significant source of pain.²²

Management of hypertonia in children with CP is challenging, and no standard approach exists. The importance of physical therapy, orthoses, and casting has been stressed by a consensus group, but these interventions can be augmented with oral medications as well as chemodenervation with BoTN, phenol, or both. Neurosurgical procedures, including ITB and selective dorsal rhizotomy (SDR), can also be considered in carefully selected populations. Less commonly, intraventricular baclofen (IVB), deep brain stimulation (DBS), and selective ventral rhizotomy can be considered in more severe or specific cases. Often, different combinations

of these treatments are used to optimize treatment of hypertonia. Treatment must be tailored to goals and may change with growth and development. Goals should be identified before initiation of any treatment. Any spasticity-reducing treatment option should translate into improved mobility, self-care, skin breakdown prevention, ease of care, or comfort and should be determined in conversation with the patient and/or family.

Oral Medications

Generalized hypertonia is often treated with oral medications. Some of the most common medications include baclofen, diazepam, dantrolene sodium, tizanidine, clonazepam, and clonidine. All of these, except dantrolene, can cause sedation because they exert their effects through the central nervous system.^{18,109} An evidence-based practice parameter looking at the pharmacologic treatment of generalized spasticity in children and adolescents with CP found limited studies and insufficient data to support or refute the use of any of these drugs on a long-term basis.¹⁸ Pediatric dosing is quite variable, and the side effects and risks of these medications limit their use.

The treatment of generalized secondary dystonia in children with CP is more challenging because it responds poorly to oral medications. Commonly used drugs include trihexyphenidyl hydrochloride (Artane), oral baclofen, benzodiazepines, gabapentin, clonidine, and levodopa-carbidopa (Sinemet). A recently published clinical practice guideline about pharmacologic and neurosurgical management of dystonia in CP by Fehlings et al. makes tiered recommendations about these agents. They suggest baclofen as first line for generalized dystonia and clonidine and/or gabapentin as second choices for severe cases.²⁸ The scientific evidence for the clinical use of trihexyphenidyl in children with CP remains equivocal.¹⁰⁶ Adverse events such as sedation, dizziness, and dry mouth are common and can limit its tolerance. ITB has been found to be more effective but at higher doses than what is required for treatment of spasticity^{2,24} (Table 48.6).

TABLE 48.6 Oral Medications for Children With Cerebral Palsy²⁴

Use	Generic Name	Dosage	Mechanism of Action	Side Effects
Spasticity/dystonia	Baclofen	Start 2.5 mg three times a day, maximum dose 80 mg/day divided three times a day in children age >8 yr	Gamma-aminobutyric acid (GABA)-B agonist	Sedation (but less than benzodiazepines), constipation
Spasticity/dystonia	Clonazepam	0.01–0.3 mg/kg/day divided twice or three times a day	GABA-A agonist	Sedation, drooling, constipation
Spasticity	Clonidine	Start at 0.05 mg/day and increase by 0.05 mg every week to a maximum of 0.3 mg/day divided three times a day	Central-acting α_2 -adrenergic agonist	Sedation, hypotension, bradycardia
Spasticity/dystonia	Dantrolene	0.5 mg/kg divided twice a day, increase every week to a maximum of 12 mg/kg/day divided four times a day	Interferes with release of calcium from sarcoplasmic reticulum in skeletal muscles	Weakness, sedation (but less than others), hepatotoxicity
Spasticity/dystonia	Diazepam	0.05–0.1 mg/kg/day divided two to four times a day	GABA-A agonist	Sedation, constipation, urinary retention

Continued

TABLE 48.6 Oral Medications for Children With Cerebral Palsy—cont'd

Use	Generic Name	Dosage	Mechanism of Action	Side Effects
Spasticity/dystonia	Gabapentin	Start 10–15 mg/kg/day divided three times a day, titrate maximum of 40 mg/kg/day divided three times a day	Unknown	Sedation, emotional lability
Dystonia	Carbidopa/levodopa	25/100: start 0.25–0.5 tablet twice a day, maximum 800 mg/day divided two or three times a day	Dopamine precursor, indirect receptor agonist	Rarely effective in cerebral palsy, more effective in genetic dystonia Gastrointestinal upset; may add extra carbidopa (Lodosyn) to alleviate side effects, sedation
Spasticity	Tizanidine	Start 1–2 mg every hour with maintenance 0.3–0.5 mg/kg/day divided three or four times a day	Central-acting α_2 -adrenergic agonists	Sedation, hypotension, hepatotoxicity
Dystonia	Trihexyphenidyl	Start 0.05 mg/kg twice daily; increase weekly by 0.05 mg/dose to a max dose of 0.75 m/kg/day ¹⁰⁶	Anticholinergic	Sedation, constipation, urinary retention, dry mouth, dyskinesias, motor tics

From Dy R, Roge D. Medical updates in management of hypertonia. *Phys Med Rehabil Clin N Am*. 2020;31(1):57-68.

Chemodenervation

Intramuscular BoNT injections and chemical alcohol (CA), including phenol and ethanol neurolysis, are used in the treatment of focal hypertonicity.³⁷ Both forms of chemodenervation (BoNT and CA) can be used at the same time to maximize the dose and number of muscles and in combination with systemic medications.⁵⁵ Typically, electrical stimulation, electromyogram (EMG), ultrasound guidance, or a combination of these is used for localization. A recent consensus statement was released discussing a tiered-based approach to consideration of BoNT use for patients with cerebral palsy.¹²⁸

BoNT is widely used in the management of CP. BoNT works by presynaptically blocking release of acetylcholine into the neuromuscular junction. Several BoNT-A products are on the market, but the only BoNT-B product is currently Myobloc. There are currently seven different serotypes of BoNT and at least five commercial preparations (e.g., Botox, Dysport, Xeomin, Myobloc, Jeuveau, Daxxify). Of the commercial preparations, more evidence is available for BoNT-A. Dosage and dilution guidelines have not been established for any of the commercial preparations, but consensus statements and systematic reviews provide recommendations.^{37,128} The child's age, weight, severity of hypertonia, and muscle localization are taken into consideration when determining dosage. The period of clinically useful relaxation is usually 12 to 16 weeks, and it is recommended that injections be spaced a minimum of 3 months apart because of concern for neutralizing antibody formation.³⁷ Adverse events related to BoNT are rare and include injection site pain, fatigue, and excessive weakness of the injected muscle. Some cases of dysphagia and urinary incontinence have been reported.¹⁸

BoNT treatment is appropriate for hypertonia in the form of spasticity and/or dystonia in all functional GMFCS levels I to V if the goal setting is adapted to the specific problems.¹²⁸ The main indication is hypertonia that hinders function and that if left untreated will become a fixed (rather than dynamic) contracture. Ideally, BoNT treatment should start at a young age when gait patterns and motor function are still flexible, allowing for gross motor function learning, including strengthening of antagonist muscle groups during the time window of tone reduction.³⁷ How-

ever, more recent discussion raises concern for the potential of progressive atrophy related to frequent recurrent injections.⁷³

Before BoNT was developed and commercialized, phenol motor point blocks were the only available form of focal chemodenervation. CAs work by proteolytically causing neurolysis. Motor point blocks, in which medicine is injected into motor points within the muscle belly, can be performed with electrical stimulator alone, while true motor nerve blocks can be performed with ultrasound guidance. CAs lyse tissue, causing injections to be far more painful, and often require sedation. Nerves that have limited sensory fibers, such as the musculocutaneous, obturator, or pectoral nerves, can be safely targeted. Injecting other nerves has a higher risk of developing dysesthesias because of sensory fiber injury. In contrast to Botox, CAs are inexpensive, and results can last between 6 and 18 months.⁵⁵

Intrathecal Baclofen Therapy

ITB is most often used to treat children with generalized spasticity or generalized moderate-to-severe dystonia.² It is a US Food and Drug Administration–approved method to treat spasticity of cerebral or spinal origin. Currently, indications for the use of ITB include tone that is thought to interfere with function or the ability to provide care, modified Ashworth scores above 3, and definable goals for spasticity reduction. ITB is delivered through a programmable pump placed in the subcutaneous tissue or myofascial layer of the abdomen connected to the intrathecal space at the desired spinal level via a tunneled catheter. This method allows the delivery of smaller doses of baclofen (micrograms) intrathecally, thus reducing the side effect profile seen with the oral form of baclofen (milligrams). The dose can be titrated to a desired therapeutic response. A screening lumbar bolus dose of baclofen can be used to evaluate medication responsiveness. The pump requires replacement every 6 to 7 years for battery replacement. For patients with more severe dystonia, intraventricular catheter placement (IVB) can be considered, though there may be increased risk for catheter migration and worsening obstructive sleep apnea compared to ITB.

A number of reports have noted that ITB reduces hypertonicity, and the effect continues long term. Reports have attempted

to evaluate ITB use and improvements in function. Some have reported improvements in positioning, activities of daily living, transfers, sleep, oral motor skills, hand use, and comfort. Motor function measured by the gross motor function measure (GMFM), pediatric evaluation of disabilities inventory, and functional independence measure for children has been reported to improve. Others have found no improvement.^{57,60}

Potential complications of ITB include infections, cerebrospinal fluid leaks, and catheter problems such as disconnection, migration, fracture, or kinking. Abrupt withdrawal is a medical emergency and may present as increased tone, spasms, diaphoresis, agitation, and pruritus. If untreated, it can progress to rhabdomyolysis and multisystem failure. Treatment includes high doses of oral baclofen as well as benzodiazepines and cyproheptadine. The latter alleviates the pruritus related to ITB withdrawal. Intrathecal baclofen through a lumbar drain can also be used in treatment of severe withdrawal. Acute complications include urinary retention, whereas constipation and weight gain tend to be more chronic complications.^{2,57} For these reasons, it is very important to verify that caregivers and family have the resources needed to comply with the longitudinal care of the ITB therapy and its potential complications.

Deep Brain Stimulation

DBS involves a stereotactic implantation of electrodes into the basal ganglia and a programmable pulse generator implanted subcutaneously in the infraclavicular region. DBS is the treatment of choice in primary dystonias. In CP, the majority of persons have generalized secondary dystonia (typically quadriplegics with GMFCS levels IV–V). ITB is the treatment of choice in this population; however, in those who do not respond adequately to ITB, DBS can be considered. The response is less than that seen in primary dystonia; some studies have reported only 30% to 33% reduction in different dystonia rating scales, such as the Burke-Fahn-Marsden scale.^{58,110}

Selective Dorsal Rhizotomy

SDR is a surgical procedure used for spasticity. The surgical technique involves single or multilevel laminectomies exposing the nerve roots and selectively cutting a percentage of the dorsal rootlets with abnormal response with the aid of electrophysiologic monitoring. For functional outcomes, the ideal candidate for SDR is a child between the ages of 3 and 8 years old with diplegic CP and predominantly spastic tone (typically GMFCS levels I–III), sufficient underlying strength, good selective motor control, and minimal contractures. Positive preoperative functional predictors for a good SDR outcome include the ability to rise from a squatted position with minimal support and a younger child's ability to crawl on hands and knees or tall kneel.¹¹⁷ The presence of ataxia or moderate to severe dystonia is a relative contraindication because both conditions could be exacerbated and unmasked after SDR. A degree of lower limb weakness can be unmasked postoperatively by reducing the spasticity, making intensive physical therapy a necessity. Children must have the cognitive and social capacity for such an intensive intervention.⁷⁷

Positive outcomes of SDR include reduced spasticity, increased ROM, improved gait pattern and kinematics, increased speed, and decreased oxygen cost, making gait more efficient.^{93,125} A meta-analysis of three randomized controlled trials evaluating the outcome of SDR in children with spastic diplegia showed that SDR plus physical therapy is more efficacious than physical therapy alone in reducing spasticity and has a small positive effect on gross motor function.⁶⁵ A study looking at 136 patients retrospectively showed that spasticity decreased substantially at the

major muscle groups: hip flexors, adductors, hamstrings, and rectus femoris. There was less reduction in the plantar flexor group, which may be functionally beneficial for joint stabilization and power production. Long-term outcomes at 5 and 20 years after SDR in children show an improvement in the GMFM and gait pattern.¹²⁵ If done at an early age, SDR can reduce bony deformity, decreasing the need for orthopedic surgery in about 35% of those young children. Long-term complications such as sensory dysfunction, bladder or bowel dysfunction, or back pain are infrequent.⁶⁸ SDR can be an alternative in nonambulatory GMFCS level IV to V children with severe spasticity if goals are to improve care and comfort. Combined dorsal and ventral rhizotomies have been performed in a few children with severe spasticity and dystonia who were not candidates for ITB.²

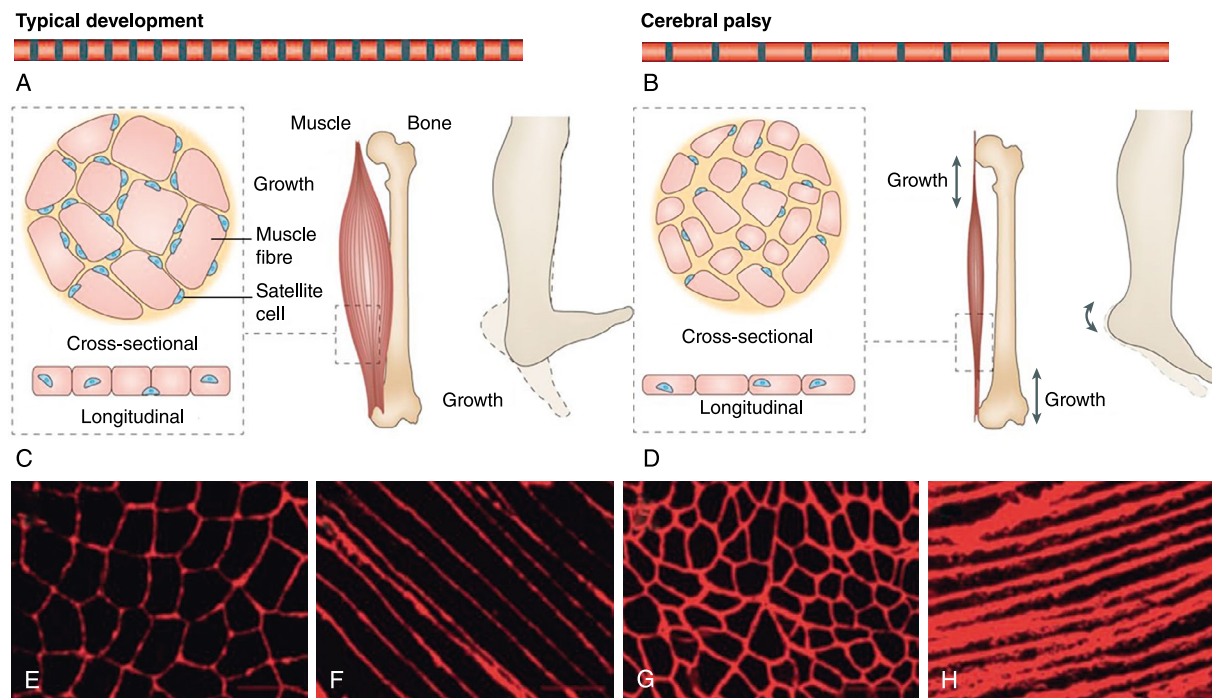
Orthopedic Management

Children with CP have abnormalities in muscle tone, strength, balance, and selective motor control as part of positive and negative features of the upper motor neuron syndrome. We now know that the muscles of children with CP are shorter and smaller and contain fibers of reduced diameter. This results in a muscle with smaller force-generating area and partially explains muscle weakness in CP. Additional intrinsic muscle differences in CP are described in Fig. 48.8.³⁸ These abnormalities in muscle tone and weakness restrict the ROM of affected joints and, when coupled with musculoskeletal growth, result in muscle contractures and associated bony deformities.^{7,38}

The introduction of the GMFCS, Functional Mobility Scale (FMS), and ICF has impacted orthopedic surgery. Surgeries designed to improve ambulation are preferred in patients at GMFCS levels I to III, whereas surgeries aimed toward improving care and comfort are performed in those with GMFCS levels IV to V. Orthopedic surgical procedures should be considered in the presence of pain, contractures that hinder function or interfere with hygiene, subluxation/dislocation of joints (most commonly the hips), and rotational problems or lever arm dysfunction, causing gait problems. There are four major types of orthopedic surgeries in CP: muscle releases and lengthening, tendon transfers, osteotomies, and arthrodesis.¹⁷ The timing of surgical intervention is determined by central nervous system maturation, ambulation potential, and the rate at which the deformity is developing. Graham et al. have suggested delaying any orthopedic surgical intervention until ages 7 to 9 years because of an elevated risk of recurrence unless there is evidence of hip subluxation. Before this age, the focus should be therapies along with tone management.^{37,38} Remember that orthopedic surgery in and of itself has no effect on motor control, strength, or muscle tone. The brief reduction in hypertonia is the direct result of the temporary decrease in muscle tension that involves the Golgi tendon apparatus and the muscle spindle. It is therefore of great importance to use a comprehensive team that addresses tone management presurgery and postsurgery.¹⁷

Hips

It is estimated one in three children with CP will have hip displacement. The highest risk is in the nonambulatory population, as high as 90% in GMFCS level V, but it does not correlate with motor type.^{38,46} The pathophysiology relates to the muscle imbalance between the strong hip adductors/flexors and the weak hip abductors/extensors. Another factor involved in the pathophysiology is the lack of weight bearing through the hip joint in nonam-



• **Fig. 48.8** Structural changes observed in muscle of children with cerebral palsy compared with typically developing children. (A, B) Schematic representation of the long sarcomere lengths observed in children with cerebral palsy (part b) compared with the shorter sarcomere lengths observed in typically developing children (part a). This dramatic sarcomere length difference is observed even though the overall fascicle lengths of the two muscles are nearly identical. (C, D) Schematic representation of the growth in muscle from typically developing children as bone length increases. Sarcomeres are added in series as the bone grows, and therefore the ankle retains full range of motion (part c, right image). However, in a child with cerebral palsy, we hypothesize that sarcomeres are not added in series and thus, as the bone grows, the ankle is forced into plantar flexion (part d, right image). In the panels on the left, the microscopic appearance of muscle is shown. In cross-sections, muscle fibres from typically developing children are larger compared with fibres of children with cerebral palsy. In addition, in longitudinal sections, fibres from typically developing children have more satellite cells compared with children with cerebral palsy. (E–H) Immunohistochemical staining of human muscle for laminin, one of the constituents of the extracellular matrix (ECM). Part e and part g are cross-sections, whereas part f and part h are longitudinal sections. Note the increased amount of ECM in muscle from children with cerebral palsy (part g and part h). (From Graham HK, Rosenbaum P, Paneth N, et al. Cerebral palsy. *Nat Rev Dis Primers*. 2016;2:15082.)

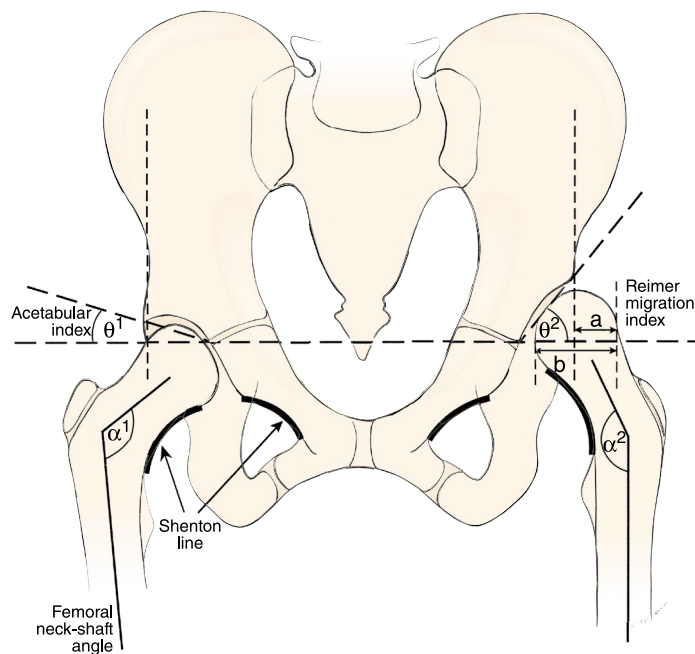
bulatory children because axial weight bearing serves to deepen the acetabulum and induces a progressive varus angle of the femoral neck with respect to the shaft. The osseous deformities of the hip that result include persistence of femoral anteversion, acetabular dysplasia, and coxa valga resulting from a lack of axial weight bearing. This in conjunction with muscle imbalance causes the head of the femur to gradually migrate out of the socket, thus compromising the hip joint integrity.⁴⁶

Soft tissue surgeries used to counteract hip subluxation often include a combination of hip adductor tenotomy, psoas recession, iliopsoas tenotomy, or medial hamstring lengthening and are performed after the hip has demonstrated subluxation. Bony surgeries include periacetabular osteotomies to improve coverage of the femoral head, proximal varus, and external rotation osteotomies of the femur to address the angular and rotational deformities, respectively.^{46,85} More recently, guided growth intervention to the proximal femur in children younger than 5 years has surged as a feasible alternative with promising outcomes.

There is limited evidence that seating and orthotic devices can prevent hip disease in nonambulators. Repeated injections of Botox combined with an abduction brace did not prevent progressive subluxation in a large, randomized clinical trial of

3-year duration. There was a delay in progression to surgery but no difference in long-term outcomes.⁴⁶

Hip surveillance programs for early detection have been implemented in many countries. Examples of such include the Pediatric Orthopedic Society of North America Hip Surveillance Guidelines¹¹⁵ and Australasian CP Hip Surveillance Guidelines. The long-term goal is to avoid the sequelae of late hip dislocation, which may include pain, fixed deformity, loss of function, and impaired quality of life.⁴⁶ These programs recommend annual or semiannual surveillance films and clinical evaluation according to GMFCS level and progression of migration percentage.^{20,46,115} Monitoring the acetabular index and migration percentage are the most effective parameters. Children with a migration percentage greater than 30% or an acetabular index over 30 degrees are likely to need further treatment. A progression of the migration percentage by more than 7% per annum requires careful monitoring (Fig. 48.9).³⁶ As with any surgical intervention, goals must be clearly established. For the ambulatory child with CP, the goal of preserved ambulation requires a contained and stable hip. In the nonambulatory child with CP, the goals include prevention of dislocation, maintenance of sitting balance, facilitation of hygiene, prevention of pain, prevention of skin breakdown, and prevention of progressive scoliosis.³⁶



• **Fig. 48.9** Radiographic parameters for the evaluation of hips in the setting of spasticity. (Redrawn from Herring JA, ed. *Tachdjian's Pediatric Orthopaedics*. 6th ed. Saunders; 2022.)

The natural history of the severely subluxed hip has been studied with variable conclusions. The relationship between hip pain and subluxation or dislocation is highly variable, but it has reached above 80% in some studies.⁴⁶ Although a subset of children with neuromuscular dysplasia has hip pain, the relationship between radiographic and clinical findings remains elusive. For those with established pain attributed to hip dislocation, deformity of the femoral head, and breakdown of articular cartilage, temporary relief with intraarticular steroid injections can be therapeutic and diagnostic. Salvage procedures to include proximal femoral resection, osteotomies to redirect the femoral head, and hip arthroplasty have all been described.⁸⁵

There are several population-based studies that show hip surveillance improves the outcome of hip pathology in CP, favoring preventive and reconstructive procedures over salvage surgery.⁴⁶

Lower Limb

In CP, positive and negative features of the upper motor neuron syndrome, delays in gross motor skill acquisition, and atypical gait patterns lead to further deformity. A diplegic sagittal gait pattern classification has been developed (Fig. 48.10), which helps define gait pathology, potential causes, and treatment strategies.⁸⁹ Scissoring gait or excessive hip adduction and stiff knee gait or overactive rectus femoris are additional gait abnormalities that may coexist in this classification. Three-dimensional gait-and-motion analysis aids in surgical planning and provides valuable kinematic, kinetic, and EMG information to further describe these gait abnormalities.

Hypertonia results in muscle contractures. In CP, biarticular muscles are more commonly contracted (e.g., psoas, rectus femoris, hamstrings, gastrocnemii) than the monoarticular muscles.^{38,78} Before any orthopedic surgical intervention, tone management is recommended especially at an early age. In patients who are good candidates for ITB therapy or rhizotomy, the preference has been for those treatments to be pursued first because the need for orthopedic surgery could be minimized.⁷⁸

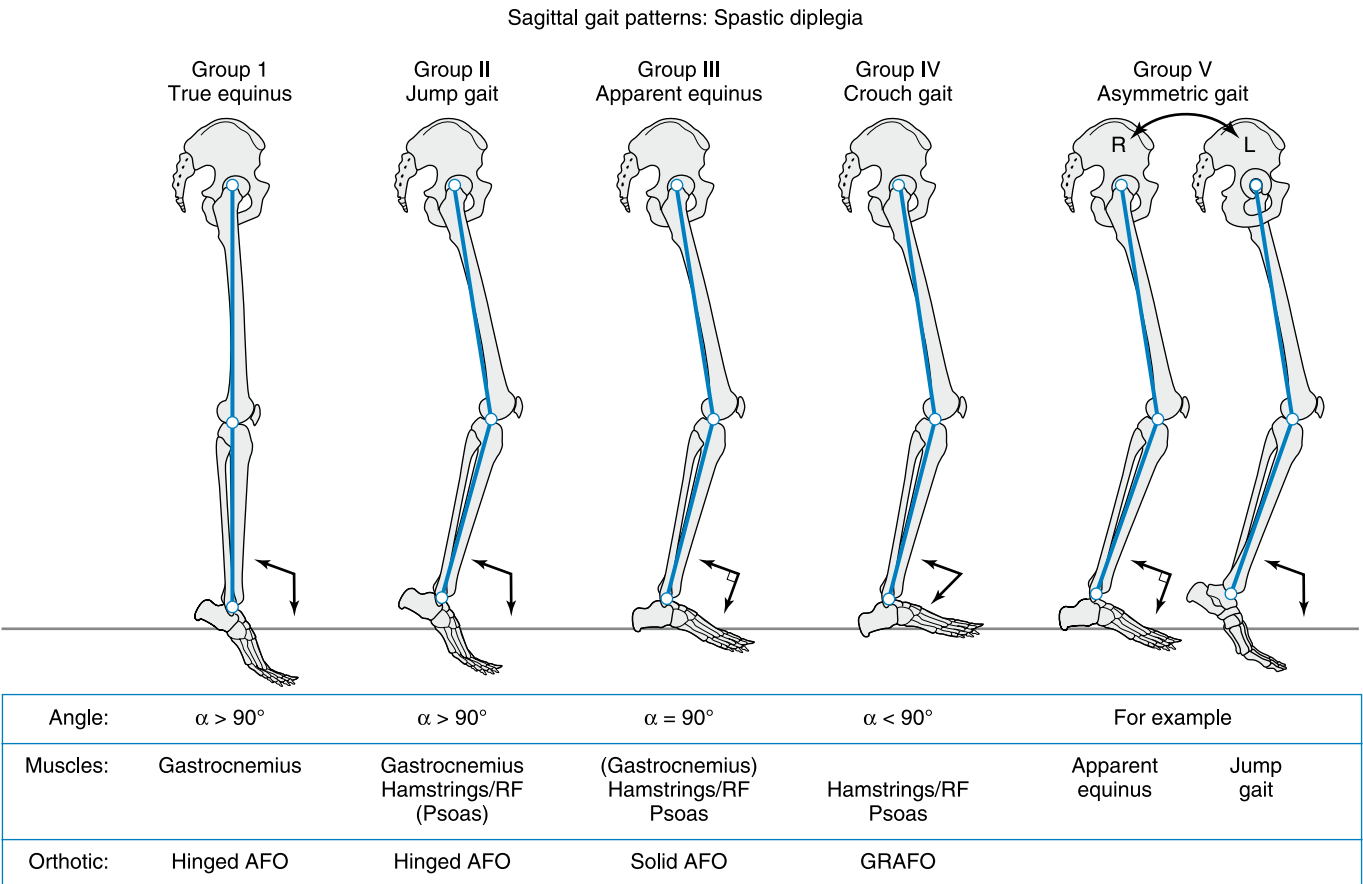
Orthopedic surgeries for contractures include lengthening and transfers. Typically, these procedures are more effective if integrated into a single-event multilevel surgery (SEMLS). SEMLS is defined as corrections of soft tissue and/or bony deformities at a minimum of two anatomic levels during a single operative event. The advantage of a SEMLS is one hospital admission and recovery period for multiple interventions.⁶¹ The goal is to improve the biomechanical impairments that limit walking activity and community participation.

Church et al. reported that on average children recover baseline walking activity after low-burden surgery relatively quickly (3 months). The recovery of baseline walking activity after high-burden surgery requires an average of 13 months.¹⁵ Patient factors such as age, GMFCS level, and rehabilitation protocols have an impact on this timeline.

Abnormal long-bone torsions (lever arm dysfunction), such as persistent femoral anteversion, are also a result of abnormal muscle forces and delays in walking. Another less common example that is acquired later in development is tibial torsion. If left untreated, lever arm dysfunction will progress to further malalignment of the bone and unwanted gait compensations (Table 48.7).³³ In the nonambulatory cohort, goals of tenotomies include facilitating sitting (hamstrings) and perineal care (adductors), maintaining braceable feet (Achilles tendon, gastrocnemius-soleus) to promote weight bearing, and preventing further hip migration (psoas and adductors) with or without bony surgery.

In CP, the development of the foot typically is not complete until the child is 7 to 8 years old.

In children with CP, spasticity at the ankle plantar flexors usually peaks at around age 5 years often with equinus contractures by age 5 to 8 years that worsen if untreated. Planovalgus foot deformities are common in young children less than age 5 years with most improving toward a normal foot posture by age 5 to 7 years, then plateauing, and becoming more severe with adolescent growth and weight gain. Varus foot deformities tend to persist in hemiparetic CP. However, in those with diplegic CP, they usually overcorrect into planovalgus without intervention.¹¹³ A host of soft tissue and bony procedures are described to address



• **Fig. 48.10** Sagittal gait patterns: spastic diplegia. AFO, Ankle-foot orthosis; GRAFO, ground reaction ankle-foot orthosis; RF, rectus femoris. (Redrawn from Rodda JM, Graham HK, Carson L, et al. Sagittal gait patterns in spastic diplegia. *J Bone Joint Surg Br.* 2004;86:251-258.)

TABLE 48.7 Gait Deviations^{7,33,78,89}

Gait Deviation	Description	Causes	Surgical Treatment	Comments
Scissoring gait	Excessive lower extremity adduction	Femoral anteversion Medial hamstring spasticity Hip adductor spasticity	Adductor tenotomy or myotomy ± hamstring lengthening	May coexist with crouch and jump gait
Jump gait	Excessive hip and knee flexion and ankle equinus	Multifactorial: hamstring, psoas, rectus femoris, and gastrocnemius contractures Weakness Bony rotational deformities (lever arm dysfunction [LAD])	Usually involves extension osteotomies of the femur along with patellar tendon advancements	Most common diplegic gait pattern in young children Jump gait may be a precursor to crouch gait
Equinus	Gastrocnemius contracture	Spasticity of the gastrocnemius muscles	Gastrocnemius recession. In hemiplegics: Baker lengthening or Achilles tendon lengthening	A plantar flexion knee extension couple can contribute to compensatory knee hyperextension (“recurvatum”) Rarely the soleus needs to be addressed except in hemiplegics Overlengthening (especially the soleus) may weaken the muscle and contribute to other gait abnormalities such as crouch

TABLE 48.7 Gait Deviations—cont'd

Gait Deviation	Description	Causes	Surgical Treatment	Comments
Crouch gait	Variable pelvis position, excessive hip and knee flexion and ankle dorsiflexion	Multifactorial: hamstring, psoas, rectus femoris, and gastrocnemius contractures and weakness Bony rotational deformities (LAD, including foot deformity)	Usually involves extension osteotomies of the femur along with patellar tendon advancements $\pm$ psoas lengthening depending on pelvis position	Most common diplegic gait pattern in older children Progressive and may lead to decline in functional independence May be iatrogenic overlengthening of muscles
Stiff knee gait	Decreased knee flexion range of motion throughout the gait cycle	Overactive rectus femoris during swing phase	Rectus femoris transfer	Commonly observed in hemiplegics

the musculoskeletal deformities of the foot and ankle in CP with the goal of obtaining a plantigrade, braceable foot and stable base of support for standing and gait.⁸⁵

Upper Limb

Tone management and orthopedic intervention for the upper limb in CP follow a similar treatment algorithm to that of the lower limb.⁸⁵ Conservative treatment with therapy, splinting (functional and passive orthoses), casting, oral medications, chemodenervation, and neurolysis are used to preserve ROM and delay muscle contractures. Upper limb surgical procedures are recommended to improve function, ease of care in patients with severe contractures, and sometimes cosmesis. Common functional impairments include deficiencies in sensation, pinch, grasp/release, and reach. The pattern of joint contractures usually involves flexor synergy: shoulder internal rotation, elbow flexion, forearm pronation, wrist and finger flexion, thumb-in-palm deformity, and swan-neck deformities. The ideal candidate for surgical intervention would be a motivated child with volitional use of the hand, spasticity without fixed contractures, and reasonable sensory function, including stereognosis, proprioception, and light touch. Patients with significant athetoid movements and dystonia often do not benefit from upper limb surgery to improve function given the variability of the outcomes.¹²⁹ In this cohort, bony fusions and arthrodesis are offered to stabilize joints. Regarding surgical timing, grip strength and manual dexterity improve with age and maturity. Although surgery before age 8 years may lead to better use of the corrected hand, transfers may work better after age 6 years, when children can participate in postoperative therapy. Tendon transfers across an open growth plate must allow for increased tension with skeletal growth to avoid deformity.¹³ Preoperative evaluation should include a detailed physical examination (ROM, strength, and tone), use of a functional classification system, and standardized tests that evaluate proprioception, stereognosis, and two-point discrimination. Motion analysis and computer modeling of the upper limb are increasingly being used to aid in surgical planning, much like three-dimensional gait analysis for the lower limbs (Table 48.8).⁸⁵

Spine

The reported incidence of scoliosis in CP is estimated to be 21% to 76%. A recent population study by Willoughby et al. found that the scoliotic severity (curves >40 degrees Cobb angle) inversely

correlated with functional and ambulatory status, GMFCS, and MACS level and with a higher prevalence in those with mixed tone and dystonia.¹³² It also found that these curves continued to progress into early adulthood.

Scoliotic curves have been characterized and divided into two groups:

Group 1: Curves are single thoracic or double thoracic and lumbar curves with level pelvis. This type of curve is typically associated with ambulatory patients (GMFCS II–III).⁴⁹

Group 2: Curves are long thoracolumbar or C-shaped curves with associated pelvic obliquity, typically associated with nonambulatory patients.⁴⁹

The natural history of curve progression has shown that onset occurs from 3 to 10 years of age in those with GMFCS level IV to V, with rapid progression during the adolescent growth spurt. Bracing traditionally has a limited role in decreasing curve progression.⁴⁹ A physical examination should be performed every 6 to 12 months, and a radiograph is obtained if a curve is detected. Observation is warranted for flexible curves less than 40 degrees that do not compromise sitting balance. In most cases of scoliosis in CP, spinal instrumentation and fusion are considered because of significant curve progression, loss of sitting balance, and comfortable function.⁴⁹ Ideally, this can be delayed in children with flexible deformities until they approach skeletal maturity. Surgery with complete spinal fusion in young children can impair growth of the spine and thorax, affecting respiratory development and function.³⁹ Growing rods (e.g., MAGEC rods) can be used to correct the deformity while allowing for growth. It is important to note that surgical complications in spine fusion are high; therefore goal setting, patient selection, and preoperative preparation are of utmost importance.

Transition to Adulthood and Aging With Cerebral Palsy

Preparing for transition begins early in the care of the individual with CP. A child must be encouraged to become as independent as possible and to separate if appropriate from caregivers in a developmentally appropriate fashion. Throughout the course of care for the child, the medical team is responsible for providing anticipatory guidance and education. Guidelines for the care of

TABLE 48.8 Operative Interventions of the Upper Extremity in Cerebral Palsy

Joint	Aim	Options
Shoulder	Joint stabilization Improve external rotation Improve internal rotation	Fusion, capsular reconstructions Lengthen pectoralis major/subscapularis; transfer LD and/or teres major; humeral osteotomy Lengthen/release infraspinatus/teres minor
Elbow	Joint stabilization Improve extension	Fusion Lengthen biceps brachii/brachialis; BR release; flexor/pronator mass release (slide); capsulotomy
Forearm	Improve supination	Reroute, lengthen, or release PT; radius/ulna osteotomy; flexor-pronator release (slide)
Wrist	Stabilization Improve extension	Fusion Flexor tendon release; proximal row carpectomy; ECU transfer; FCU transfer to ECRB/ECRL/EDC
Thumb	Stabilization Improve extension Improve abduction	Volar plate arthroplasty: MCP fusion Release/lengthen FPL; reinforce EPL Release adductor pollicis; reinforce APL; EPL rerouting
Fingers	Flexion deformity Swan-neck deformity	FDS to EDC transfer flexor/pronator release (slide); FDS/FDP lengthening; FDS to FDP transfer PIP joint tenodesis; central slip tenotomy; intrinsic origin release

APL, Abductor pollicis longus; BR, brachioradialis; ECRB, extensor carpi radialis brevis; ECRL, extensor carpi radialis longus; ECU, extensor carpi ulnaris; EDC, extensor digitorum communis; EPB, extensor pollicis brevis; EPL, extensor pollicis longus; FCU, flexor carpi ulnaris; FDP, flexor digitorum profundus; FDS, flexor digitorum superficialis; FPL, flexor pollicis longus; LD, latissimus dorsi; MCP, metacarpophalangeal; PIP, proximal interphalangeal; PT, pronator teres.

From Koman LA, Sarikiotis T, Smith BP. Surgery of the upper extremity in cerebral palsy. *Orthop Clin North Am.* 2010;41:519-529.

children and adolescents with CP exist and can help primary care physicians with ensuring comprehensive assessments of medical and psychosocial issues encountered in the CP population, direction on management of associated conditions, educational and community support needs, and suggested referrals to specialists for specific issues.^{76,97} Conversations should be fostered with families to encourage patient autonomy and avoid infantilization when appropriate and may include challenging conversations about sexual and mental health.

Therapy services for older individuals with CP are indicated for specific goals that will enhance functional independence or exercise capacity. For some individuals, independent living with employment is very realistic. Others may thrive in a supported environment with some daily assistance but not require constant supervision or care. Children with significant cognitive and physical delays may require long-term support but may enjoy opportunities for socialization with peers other than their primary caregivers. Children with disabilities may need encouragement and self-advocacy training. Coping skills and self-reliance are usually acquired through experiences leading up to the point of transition and cannot be simply taught once an individual reaches adulthood.

Transition of health care occurs in phases and requires preparation, readiness, and handoffs. Care delivery models often differ in the pediatric and adult realms, and moving to a new group of providers can be anxiety provoking for many. Teens benefit from having advocates or champions to help with the transition process. Mindful practitioners collaborate with interested parties, including families, schools, and community agencies, to provide comprehensive guidance.¹²⁰ Patient counseling needs to include reminders about preventative health measures and screening. Health conditions affecting the general population certainly will occur in adults with CP also. There are also specific conditions that occur more frequently or earlier in the adult CP population as the result of aging and chronic alterations in biomechanics, such as cervical myelopathy and early development of degenerative joint disease.

Challenges exist for the CP population as a whole associated with aging. Research shows that adults with CP have a higher risk

of neurologic conditions such as stroke, myelopathy, and dementia.¹¹⁶ Musculoskeletal conditions may worsen, and energy consumption may increase for tasks previously accomplished easily. Adults with CP may experience chronic pain and associated fatigue and depression.^{116,132} Many adults with CP will worsen by one GMFCS level as they age; many lose their ability to ambulate independently.¹¹⁶ It is important to avoid a downward spiral when an exacerbation of a health issue or painful condition limits activity and functional decline ensues. With a shift in focus in measuring participation and quality of life indicators, the roles of physical fitness and strengthening in adults with CP are being investigated in greater detail.¹²⁴

Summary

In summary, caring for a person with CP is a lifelong commitment requiring ongoing medical management, therapies and support systems. A team of specialists which include physiatrists works with the person and family to address specific needs. Care should be family centered and goal directed.

Ongoing monitoring of health, function and addressing potential complications across the life span is crucial.

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